

SHISHIR SINGH

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Computer Science undergraduate with hands-on experience in AI engineering, Python, databases, Linux systems programming, Bash automation, Computer Networks, Operating Systems, and C/C++.

EDUCATION

Vellore Institute of Technology

B.Tech. Computer Science and Engineering (CGPA: 8.1)

Bhopal, MP

2023 - 2027

EXPERIENCE

Summer Intern

Ciena India Pvt. Ltd.

May 2026 - July 2026

Gurugram, HR

- Automated network hardware monitoring across 25 switches using Bash, Telnet, and cron, enabling port-level connectivity validation and reducing manual diagnostic effort.
- Enhanced process communication and synchronization using Linux IPC mechanisms, including pipes, shared memory, sockets, and message queues.
- Troubleshoot and debugged Linux networking and embedded software workflows using shell utilities, logs, SSH, and networking tools while collaborating with engineers on access-network systems.

PROJECTS

LawBot | *Python, LangChain, Llama 3, Pinecone, FastAPI, Whisper, RAG, Hybrid Search, SHAP*

- Developed an AI-powered legal assistant using RAG + LLM to help users understand legal rights, file grievances, and access government schemes, addressing the challenge of navigating complex, unstructured legal documents.
- Implemented a hybrid retrieval and reranking pipeline with source-grounded citations and verified web research, integrated a Whisper-based voice interface, and deployed via FastAPI for real-time, context-aware responses.

TCP Client-Server File Transfer System | *Python, TCP sockets, Wireshark, tcpdump*

- Developed a client-server application in Python using TCP sockets for reliable file transfer across networked systems; implemented connection establishment, packet validation, timeout handling, retransmission logic, and error recovery mechanisms.
- Designed networking workflows following TCP/IP communication principles and socket programming concepts; analyzed network traffic using Wireshark and tcpdump to debug protocol behavior and optimize communication reliability.

SKILLS

Technical Skills

Backend & APIs

Databases & ORMs

Machine Learning & AI

Data Analysis & Visualization

Tools & Platforms

C++ (Data Structures & Algorithms), Python, JavaScript, Bash

Flask, FastAPI, REST APIs

PostgreSQL, MySQL, MongoDB

LangChain, LangGraph, Scikit-learn, LLMs, RAG

Pandas, NumPy, Power BI

Docker, AWS, Git, Postman, Linux

ACHIEVEMENTS

- Accepted 300+ Data Structures and Algorithms problems on [Codeforces](#) and [LeetCode](#); achieved Codeforces Pupil rating of 1214.
- Successfully cleared [GenW.AI Level 1](#) at Deloitte Hackspllosion 2026 Hackathon.
- Oracle Agentic AI Foundations Associate